

DSBDAL Assignment 14

Mini Project — Covid Vaccine Statewise Analysis

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Subject	Data Science & Big Data Analytics Lab (DSBDAL)
Assignment	Assignment 14 — Mini Project 1
Dataset	covid_vaccine_statewise.csv
Language	Python 3.10 Pandas · Matplotlib · Seaborn
Platform	Google Colaboratory

A comprehensive analysis of India's COVID-19 state-wise vaccination data, examining dose administration, gender-based vaccination patterns, and identifying top-performing states.

1. Introduction & Objectives

1.1 Background

The COVID-19 pandemic necessitated one of the largest vaccination drives in human history. India rolled out its vaccination programme in phases starting January 2021, administering multiple vaccines — primarily Covishield (Oxford-AstraZeneca), Covaxin (Bharat Biotech), and Sputnik V — across all 28 states and 8 Union Territories.

Understanding the distribution, pace, and demographic spread of vaccinations is critical for public-health planning. This project analyses the **covid_vaccine_statewise.csv** dataset, which contains cumulative daily vaccination figures for each state and the national aggregate from January 2021 onwards.

1.2 Objectives

- Load and inspect the raw vaccination dataset.
- Perform data preprocessing — clean column names, convert types, filter aggregates.
- Extract key vaccination statistics per state (First Dose, Second Dose).
- Quantify gender-wise vaccination counts (Male vs. Female).
- Create informative visualisations to highlight top-performing states.

2. Dataset Description

2.1 Source & Overview

The dataset (*covid_vaccine_statewise.csv*) is sourced from India's Ministry of Health vaccination portal. Each row represents a single day's cumulative figures for a given state/UT.

Attribute	Detail
Total Records	7,845 rows
Total Columns	24 columns
Date Range	16 Jan 2021 — latest available
Coverage	All Indian States + UTs + National aggregate
Memory Usage	~1.4 MB

2.2 Column Glossary

Column Name	Description
Updated On	Date of the record (DD/MM/YYYY)
State	State or Union Territory name
Total Doses Administered	Cumulative total doses given up to this date
Sessions	Number of vaccination sessions conducted
Sites	Number of active vaccination sites

Column Name	Description
First Dose Administered	Cumulative 1st dose count
Second Dose Administered	Cumulative 2nd dose count
Male (Doses Administered)	Total doses given to males
Female (Doses Administered)	Total doses given to females
Transgender (Doses Administered)	Total doses given to transgender persons
Covaxin (Doses Administered)	Doses of Covaxin administered
CoviShield (Doses Administered)	Doses of CoviShield administered
Sputnik V (Doses Administered)	Doses of Sputnik V administered
AEFI	Adverse Events Following Immunisation count
18-44 / 45-60 / 60+ (Doses)	Age-group wise dose counts
18-44 / 45-60 / 60+ (Individuals)	Age-group wise unique individual counts
Male/Female/Transgender (Indiv.)	Gender-wise unique individual counts
Total Individuals Vaccinated	Total unique individuals with at least 1 dose

2.3 Missing Values Summary

Several columns have significant missing values, particularly age-group breakdowns and individual-level gender counts, which were only introduced later in the rollout.

Column Group	Missing Count	% Missing
Updated On / State	0	Complete
Total / First / Second Doses	224	~2.9 %
Male / Female (Doses)	384	~4.9 %
Sputnik V (Doses)	4,850	~61.8 %
18-44 / 45-60 / 60+ (Doses)	6,143	~78.3 %
Male/Female/Trans (Individuals)	7,685	~97.9 %

3. Libraries & Environment

The notebook runs on Python 3.10 (Google Colab) and relies on three core libraries:

Library	Version	Usage in this Project
pandas	1.x / 2.x	DataFrame creation, groupBy, aggregation, type conversion, null counting.
matplotlib	3.x	Foundation plotting library — figure sizing, titles, axis formatting.
seaborn	0.12+	Statistical visualisation on top of Matplotlib; barplot() with hue support.

Import Statement

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

4. Data Loading & Initial Inspection

4.1 Loading the Dataset

The CSV is read with **pandas.read_csv()**. The dataset contains **7,845 rows** and **24 columns**. The first five rows reveal date-indexed cumulative figures for 'India' (national aggregate) and individual states.

```
df = pd.read_csv('covid_vaccine_statewise.csv')
print('First 5 rows:')
df.head()
```

4.2 Dataset Info

df.info() shows column types and non-null counts. All columns except *Updated On* and *State* (object) are stored as **float64**.

```
print('Dataset Info:')
df.info()

# Key excerpt:
# RangeIndex: 7845 entries, 0 to 7844
# Data columns (total 24 columns):
# 0 Updated On 7845 non-null object
# 1 State 7845 non-null object
# 2 Total Doses Administered 7621 non-null float64
# ...
```

4.3 Statistical Summary

df.describe() yields 8 statistics for every numeric column. Selected highlights:

Column	Mean	Std Dev	Min	Max
Total Doses Administered	~9.2M	~37.5M	7	~513M
First Dose Administered	~7.4M	~30M	7	~400M
Second Dose Administered	~1.8M	~7.6M	0	~113M

Column	Mean	Std Dev	Min	Max
Male Doses Administered	~3.6M	~17.4M	0	~270M
Female Doses Admin.	~3.2M	~15.2M	2	~240M

4.4 Missing Values Count

`df.isnull().sum()` quantifies the data gaps. The *Male/Female/Transgender (Individuals)* columns are nearly empty (~97.9 %), as are age-group dose fields. Sputnik V data was only introduced mid-rollout.

```
df.isnull().sum()

# Selected output:
# Total Doses Administered 224
# First / Second Dose Administered 224
# Male / Female (Doses) 384
# Sputnik V (Doses) 4850
# 18-44 Years (Doses) 6143
# Male(Individuals Vaccinated) 7685
# Female(Individuals Vaccinated) 7685
```

5. Data Preprocessing

Step 1 — Strip Column Whitespace

Column names in the raw CSV contain leading/trailing spaces. `str.strip()` removes these for consistent key access.

```
df.columns = df.columns.str.strip()
```

Step 2 — Convert Numeric Types

Four key columns are explicitly coerced with `pd.to_numeric(..., errors='coerce')` to handle any stray non-numeric strings (which become NaN).

```
numeric_cols = [
    'First Dose Administered',
    'Second Dose Administered',
    'Male (Doses Administered)',
    'Female (Doses Administered)'
]
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

Step 3 — Remove National Aggregate Rows

All 212 rows where `State == 'India'` are excluded to prevent the national total from distorting state-level comparisons, leaving **7,633 records**.

```
india_rows_count = df[df['State'] == 'India'].shape[0]
print(f"India rows: {india_rows_count}") # 212

state_df = df[df['State'] != 'India']
print(f"Remaining: {state_df.shape[0]}") # 7633
```

6. Data Extraction & Key Findings

After cleaning, four key aggregations are computed. For cumulative time-series data, the **maximum** value per state represents the latest (most complete) recorded figure.

States Covered	Total Males Vaccinated	Total Females Vaccinated	Combined M + F Doses
36	270,163,622	239,518,609	509,682,231

6.1 First Dose — State-wise Maximum

The maximum 'First Dose Administered' per state is extracted via `groupby().max().astype(int)`, giving the highest cumulative 1st-dose count.

```
first_dose = state_df.groupby('State')['First Dose Administered']  
.max().astype(int)
```

Top 10 States — First Dose Administered

Rank	State	First Doses Administered
1	Uttar Pradesh	45,932,488
2	Maharashtra	35,040,812
3	Madhya Pradesh	29,723,036
4	Gujarat	28,101,222
5	Rajasthan	27,008,606
6	Karnataka	25,847,691
7	Bihar	23,350,171
8	West Bengal	23,257,417
9	Tamil Nadu	20,836,674
10	Andhra Pradesh	17,628,583

6.2 Second Dose — State-wise Maximum

Similarly, the maximum 'Second Dose Administered' per state provides the fully-vaccinated coverage baseline at the time of data collection.

```
second_dose = state_df.groupby('State')['Second Dose Administered']  
.max().astype(int)
```

Top 10 States — Second Dose Administered

Rank	State	Second Doses Administered
1	Maharashtra	12,112,554

Rank	State	Second Doses Administered
2	West Bengal	9,132,961
3	Gujarat	9,051,153
4	Uttar Pradesh	8,515,236
5	Rajasthan	8,375,056
6	Karnataka	7,432,852
7	Kerala	6,426,984
8	Andhra Pradesh	6,214,312
9	Madhya Pradesh	5,733,640
10	Tamil Nadu	4,686,034

6.3 Gender-wise Totals

National male and female totals come from the maximum value across all rows (the 'India' aggregate row before filtering).

```
total_males = int(df['Male (Doses Administered)'].max())
total_females = int(df['Female (Doses Administered)'].max())

print('Males:', total_males) # 270,163,622
print('Females:', total_females) # 239,518,609
print('Combined:', total_males + total_females) # 509,682,231
```

Metric	Count	% of Combined
Males Vaccinated	270,163,622	52.99 %
Females Vaccinated	239,518,609	47.01 %
Combined (M + F)	509,682,231	100 %

7. Visualisations

Four plots are generated to communicate the findings visually, using Seaborn on top of Matplotlib.

7.1 Top 10 States — First Dose Administered

A horizontal bar chart ranks the 10 states with the highest first-dose count using the *viridis* palette. Uttar Pradesh leads significantly (~45.9M), followed by Maharashtra (~35M) and Madhya Pradesh (~29.7M).

```
top_10_first_dose = first_dose.nlargest(10)
plt.figure(figsize=(12, 6))
sns.barplot(x=top_10_first_dose.values, y=top_10_first_dose.index,
palette='viridis', hue=top_10_first_dose.index, legend=False)
plt.title('Top 10 States by First Dose Administered')
plt.xlabel('Number of First Doses Administered')
plt.ylabel('State')
plt.tight_layout()
plt.show()
```

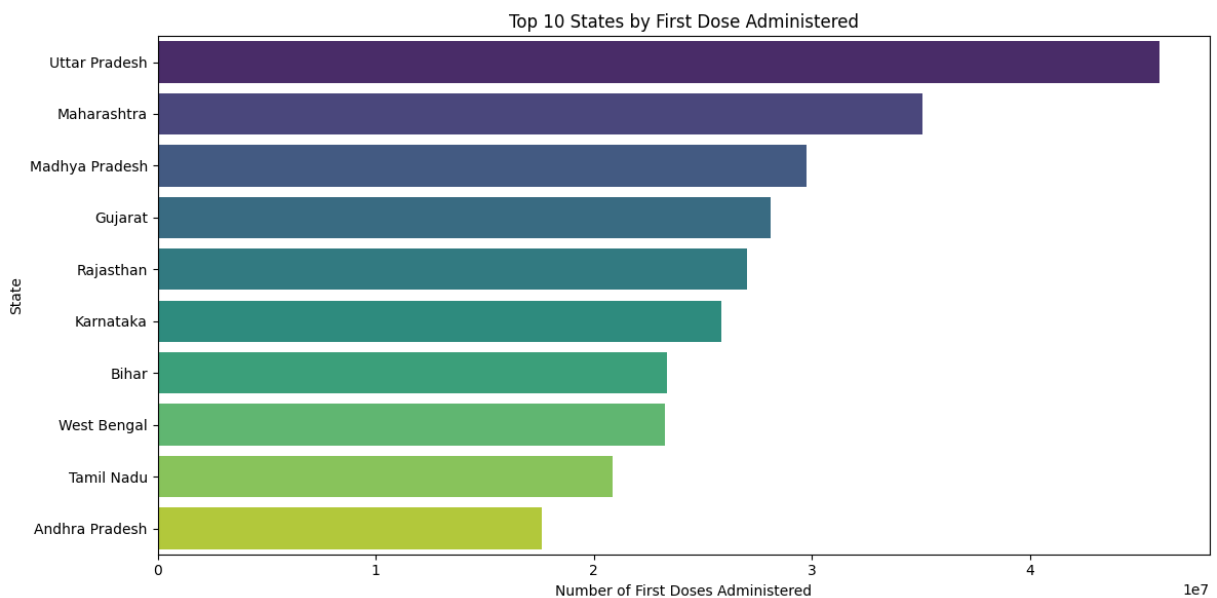


Figure 1 — Top 10 States by First Dose Administered (*viridis* palette)

7.2 Top 10 States — Second Dose Administered

A similar chart using the *magma* palette. Maharashtra leads second-dose completion (~12.1M), with West Bengal and Gujarat also ranking high. Notably, Uttar Pradesh (top in 1st doses) drops to 4th for 2nd doses.

```
top_10_second_dose = second_dose.nlargest(10)
plt.figure(figsize=(12, 6))
sns.barplot(x=top_10_second_dose.values, y=top_10_second_dose.index,
palette='magma', hue=top_10_second_dose.index, legend=False)
plt.title('Top 10 States by Second Dose Administered')
plt.xlabel('Number of Second Doses Administered')
plt.ylabel('State')
plt.tight_layout()
plt.show()
```

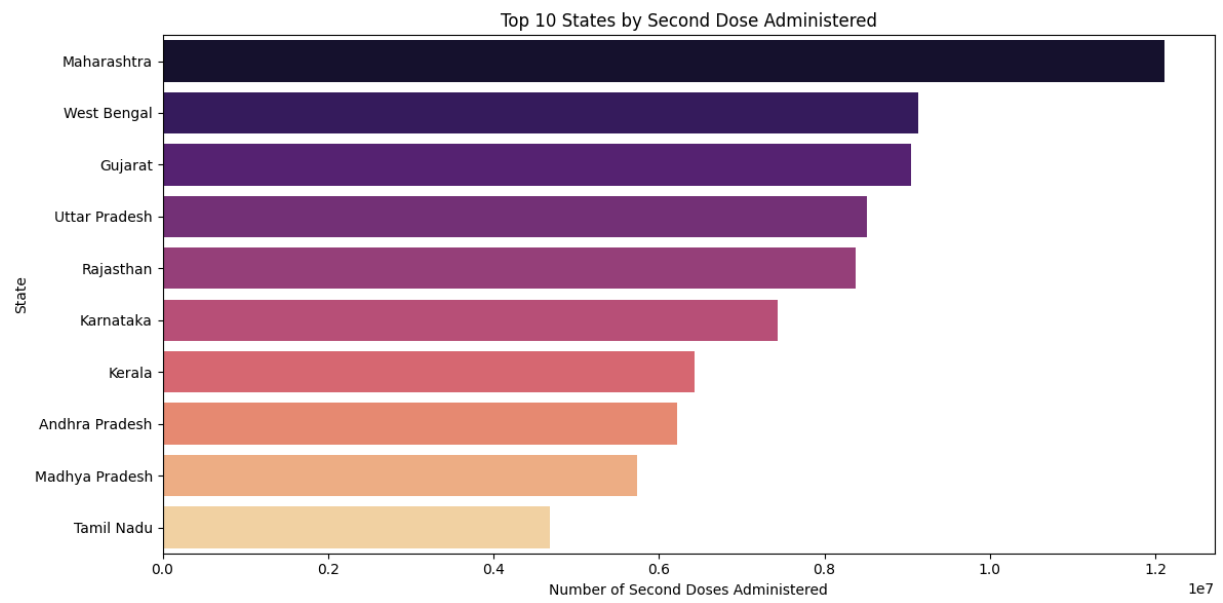


Figure 2 — Top 10 States by Second Dose Administered (magma palette)

7.3 Total Male vs. Female Vaccinations (National)

A simple two-bar chart comparing national male (~270M) and female (~239M) totals. The ~13 % gap highlights a consistent gender disparity across the vaccination drive.

```
vaccination_gender = pd.DataFrame({
    'Gender': ['Male', 'Female'],
    'Total Doses Administered': [total_males, total_females]
})
plt.figure(figsize=(8, 5))
sns.barplot(x='Gender', y='Total Doses Administered',
    data=vaccination_gender, hue='Gender', legend=False)
plt.title('Total Male vs Female Vaccinations')
plt.tight_layout()
plt.show()
```

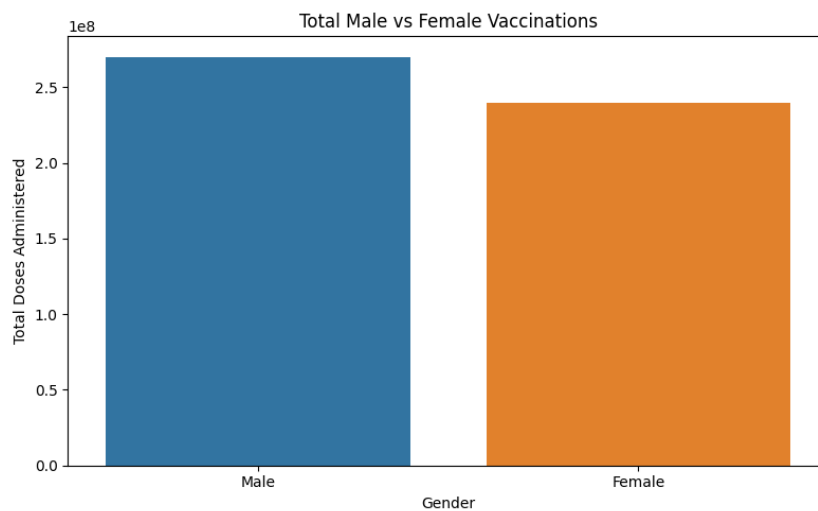


Figure 3 — National Male vs. Female Total Doses Administered

7.4 Top 10 States — Male vs. Female Vaccinations

A grouped bar chart for the top 10 states compares side-by-side male (blue) and female (brown) dose counts. Males consistently received more doses than females across all states, though the gap narrows in southern states like Tamil Nadu.

```
top_states_melted = top_states_gender_vaccination.melt(
    id_vars='State',
    value_vars=['Male (Doses Administered)',
    'Female (Doses Administered)'],
    var_name='Gender', value_name='Total Doses Administered')

top_states_melted['Gender'] = top_states_melted['Gender'].replace({
    'Male (Doses Administered)': 'Male',
    'Female (Doses Administered)': 'Female'
})

plt.figure(figsize=(14, 7))
sns.barplot(x='State', y='Total Doses Administered',
    hue='Gender', data=top_states_melted,
    palette={'Male': 'blue', 'Female': 'brown'})
plt.xticks(rotation=45, ha='right')
```

```
plt.tight_layout()
plt.show()
```

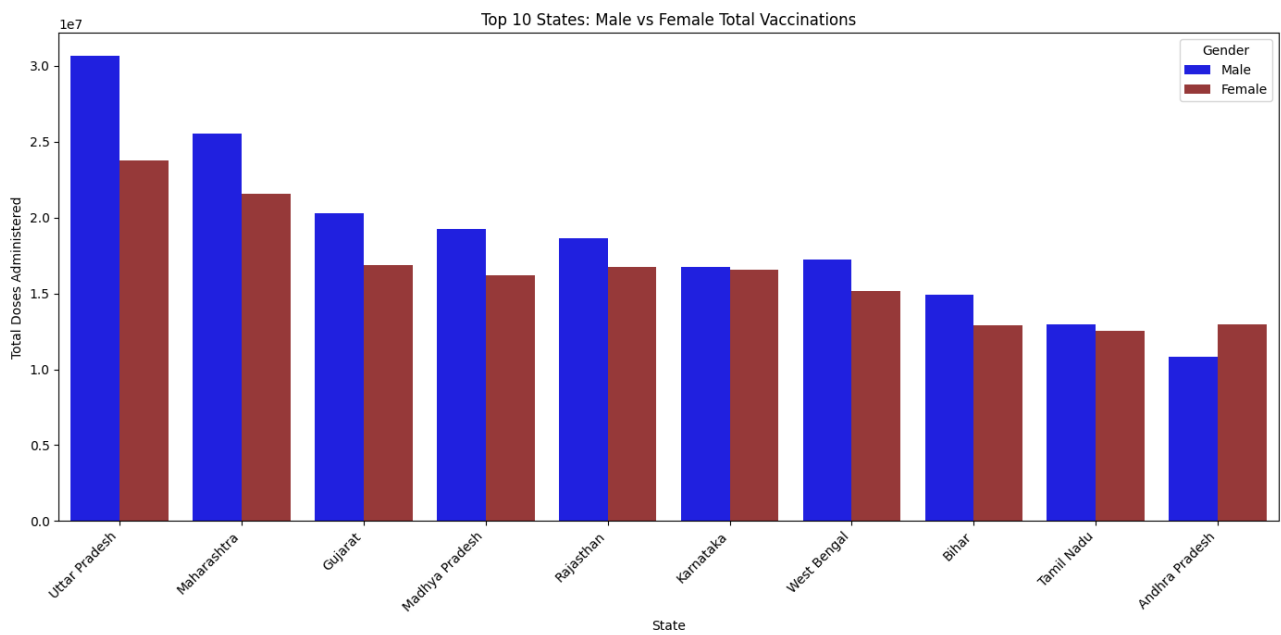


Figure 4 — Top 10 States: Male vs. Female Total Vaccinations

8. Results & Observations

First Dose Coverage

Uttar Pradesh, India's most populous state, administered the highest first doses (~45.9M). Large, populous states dominate. Smaller territories like Lakshadweep (~51K) and Ladakh (~188K) naturally record the fewest.

Second Dose Follow-through

Maharashtra leads second-dose administration (~12.1M). States with very high first-dose counts (e.g., UP: 45.9M first vs 8.5M second) show significant gaps, suggesting many individuals had not yet completed the two-dose regimen when data was collected.

Gender Disparity

Males received ~270M doses vs. ~239M for females — a ~13 % gap consistent across states. This likely reflects socio-economic and access factors. The combined 509M+ doses demonstrate the enormous scale of India's vaccination effort.

Missing Data Implications

Age-group and individual-level gender breakdowns have high missingness (>78 % for age groups, ~98 % for individual gender counts). Any analysis of these fields must account for potential bias introduced by collection gaps in the campaign's early phases.

9. Conclusion

This mini project demonstrates a complete data analysis pipeline — from raw CSV loading to insightful visualisations — on a real-world public-health dataset.

- India's vaccination drive achieved massive scale, with over 500M combined male and female doses recorded.
- Uttar Pradesh dominates first-dose distribution while Maharashtra leads second-dose completion.
- A consistent ~13 % gender gap (more males vaccinated) persists across states.
- High missingness in age and individual-count columns limits deeper demographic analysis.
- Preprocessing steps — removing the 'India' aggregate and stripping column whitespace — are critical for accurate state-level insights.